

B.E. IInd year Examination Dec 2023
IT (A&B)
3ITRC4: Digital Electronics

TIME-3 Hr

MM-60

Note: All questions are Compulsory. Attempt any two from each question.

- Q.1) A. Convert the decimal number 250.5 to base 3, base 4, base 7 and base 8. (6)
 B. What is the difference between standard and canonical form of Boolean function? Convert the following function to other canonical form. (6)

$$F(A, B, C, D) = \sum(0, 2, 6, 11, 13, 14)$$

 C. Simplify the Boolean function using tabulation method. (6)

$$F(w, x, y, z) = \sum(0, 1, 3, 7, 9, 11, 15)$$
- Q.2) A. Design a circuit that performs BCD to gray code conversion. (6)
 B. Implement full subtractor using two half subtractors and an OR gate. Implement the circuit with NAND gates also. (6)
 C. What is decoder? Design a four bit BCD to decimal decoder with proper circuit diagram. (6)
- Q.3) A. What is difference between multiplexer and demultiplexer. Implement the following function using multiplexer. (6)

$$F(A, B, C, D) = \sum(0, 1, 3, 4, 8, 9, 13, 15)$$

 B. Simplify the Boolean function using K-map (6)

$$F(A, B, C, D, E) = \sum(0, 2, 4, 6, 9, 11, 13, 15, 17, 21, 25, 27, 29, 31)$$

 C. What is race around condition and how it can be removed? (6)
- Q.4) A. Explain the working of ring counter with proper circuit diagram. What is the basic difference between ring counter and Johnson counter? (6)
 B. Write the number of clock cycles required for each of the type of shift register. Explain the working of parallel input and serial output shift register. (6)
 C. Design a counter with the following binary sequence: 0, 4, 2, 1, 6 and repeat using JK flip flop. (6)
- Q.5) A. Define the following terms regarding A to D conversion with proper example. (6)
 1. Sampling 2. Quantization 3. Encoding
 B. Explain two different methods of digital to analog conversion and which one is better method among them. (6)
 C. How many number of clock cycles are required in each of the method of A to D conversion. Which method takes least no of clock cycles for conversion and also explain its working with circuit diagram. (6)